

HOMEWORK: AI Image Explorer

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AI Image & Multimodal Models — Explore, Create, and Think Critically

Student Information

Name:

Date:

SECTION 1: Vocabulary Match (20 points)

Match each term with its definition. Write the letter in the box.

- A. AI model that creates pictures from text descriptions
- B. The process of removing noise step by step to create an image
- C. AI that works with multiple types of data (text + image + audio)
- D. AI-generated fake image or video of a real person
- E. The text description you give to an image generator
- F. How AI starts with random static and sculpts it into a picture
- G. Applying one artistic style to another image
- H. AI that can look at a photo and answer questions about it

1. Text-to-Image:

2. Diffusion:

3. Multimodal:

4. Deepfake:

5. Image Prompt:

6. Noise-to-Image:

7. Style Transfer:

8. Vision AI:

SUBMISSION: 1. Microsoft Teams 2. Copy to Ishwari Raut ma'am

Due: Next class | Total Points: 100 (+10 bonus)

SECTION 2: True or False (10 points)

Write T (True) or F (False) in each box.

1. AI image models copy real photos from the internet.
2. Diffusion works by removing noise step by step.
3. A more detailed prompt usually gives a better image.
4. Multimodal AI can only understand text, not images.
5. It is always okay to make AI images of real people.
6. AI-generated images can have mistakes like extra fingers.
7. DALL-E, Midjourney, and Stable Diffusion are text-to-image models.
8. You should always trust photos you see online.

SECTION 3: Short Answer (20 points)

Answer each question in 2-3 sentences.

1. Explain how diffusion creates an image. Use an analogy
(like sculpting, cleaning fog, or developing a photo).

2. What is the difference between a text-to-image model and a multimodal model? Give an example of each.

3. How is prompting for images different from prompting for text?
What details should you include in an image prompt?

SECTION 4: Applied Scenarios (30 points)

Scenario A: Prompt Improvement

Your friend typed: 'make a cat picture'. Rewrite this as a GREAT image prompt that includes subject, style, lighting, and mood.

Your improved prompt:

Scenario B: Real vs. Fake

You see a news photo of a celebrity doing something shocking.

List 3 steps you would take to check if it's real or AI-generated.

Scenario C: Right Tool for the Job

For each task, write whether you need: text-to-image, multimodal, or a text LLM.

Create a poster for a school event:

Ask AI to describe what's in a photo:

Write an essay about climate change:

Turn a sketch into a realistic image:

Scenario D: Ethical Dilemma

A friend wants to create a funny AI image of their teacher.

Is this okay? What would you tell them? (3 sentences)

SECTION 5: Creative Challenge (20 points)

Your Mission: Build a Prompt Collection

Create 3 image prompts for different purposes. For each one, write the prompt, say what style and mood you chose, then actually try it on an AI tool and describe the result.

Prompt 1 — For a school project or presentation:

Prompt:

Style/Mood:

Result (describe or rate 1-5):

Prompt 2 — Pure creativity (fantasy, sci-fi, anything!):

Prompt:

Style/Mood:

Result:

Prompt 3 — A real-world use (logo, room design, outfit idea):

Prompt:

Style/Mood:

Result:

BONUS: AI Art Gallery (+10 points)

Pick your **BEST** generated image from this homework. Describe it and explain what made the prompt work so well. What would you change?